Multi-Physics Process Simulation of Static Magnetic Fields in High Power Laser Beam Welding of Aluminum

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Abstract: A three-dimensional turbulent steady state numerical model was used to investigate the influence of a stationary magnetic field during partial penetration high power laser beam keyhole welding of thick aluminum parts. COMSOL Multiphysics was used to calculate the three-dimensional heat transfer, fluid dynamics and electromagnetic field equations. Thermo-capillary (Marangoni) convection at the upper weld pool surface, natural convection due to gravity and latent heat of solid-liquid phase transition were taken into account. It shows that the application of steady magnetic fields produces a braking Lorentz force in the melt based on the Hartmann effect. The flow pattern in the weld pool and also the temperature distribution and associated weld pool geometry thus change significantly. Convective flows in the melt can effectively be suppressed and the influence of thermo-capillary flow is diminished to a thin surface layer.

Keywords: electromagnetic weld pool control, laser beam welding, Lorentz force, Marangoni convection, buoyancy

1. Introduction

The high-power laser beam welding enforces the keyhole-mode welding mode, where a small amount of metal in the region of the highest laser intensity vaporizes and a vertical cavity in the work-piece called keyhole is built.

Well-known issues in deep penetration welding of aluminum are the highly dynamical flow behavior of the melt due to its low viscosity. Combined with the high heat conductivity the resulting weld pool is very wide. The weld surface becomes unstable and spattering and melt ejections occur [1].

The largest size of the weld bead occurs near the surfaces being supported by thermo-capillary convection caused by temperature-dependent surface tension variations. The resulting weld cross sections show a typical wineglass-shape

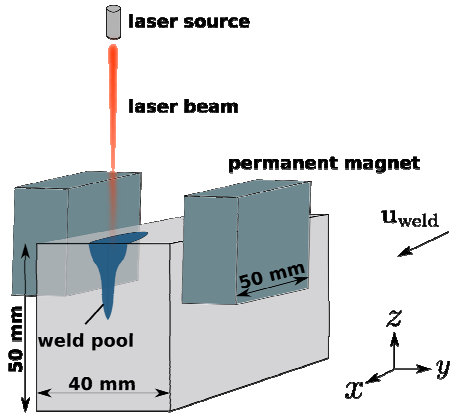


Figure 1. Experimental set-up of the laser beam welding of aluminum with steady magnetic field applied perpendicular to the welding direction.

with a strong curvature of the solidification front leading to heavy stresses in the work-piece and comparable large distortions after cooling down [2].

As the weld bead becomes larger with deeper penetration of the laser, that issue is of particular importance for high power deep penetration laser beam welding [3].

The aim of this investigation is to study the influence of static magnetic fields perpendicular to the welding direction and their effect on the liquid metal flow in the weld pool (Figure 1).

The use of magnetic fields is quite common in many industrial applications to obtain different goals as grain refinement, surface stabilization, stirring, for example in casting of metal melts as well as in arc and beam welding [4 – 11], e.g.

The present investigation deals with the application of a steady state magnetic field to a 20 mm partial penetration weld in a 50 mm thick aluminum plate. The permanent magnets were perpendicularly aligned to the welding direction. The magnetic flux density was increased up to a value of 2 T. The aim is to clarify the extent of

influence of liquid metal flow deceleration for high Hartmann numbers.

An experimental investigation of the Hartmann effect in CO_2 laser beam welding is presented in [12] showing the interaction of a stationary magnetic field with the flow of liquid metal in the weld pool. A smoothing of the weld seam and also the prevention of the humping phenomena, a periodic sagging of the melt, was reported depending on the polarity of the applied field. However, the Hartmann number was small and all effects observed relate to additional electric currents in the work-piece that existed due to the CO_2 laser plasma. Therefore, a disc laser was used in the experimental part of this investigation.

2. Physical Background

The flow of electrically conducting liquid perpendicular to an applied magnetic field causes a contribution to the electric current density according to the generalized Ohm's law

$$\boldsymbol{j} = \sigma\left(\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B}\right),$$

where $\boldsymbol{j}$ is the electric current density, σ is the electric conductivity, $\boldsymbol{E}$ is the electric field, $\boldsymbol{u}$ is the liquid metal velocity and $\boldsymbol{B}$ is the magnetic flux density.

The electric currents together with the external applied magnetic field build a Lorentz force field that has a component that is directed against the original melt flow direction independently of the polarity of the magnetic field. This is called Hartmann effect.

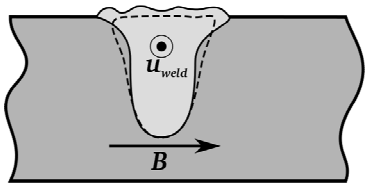


Figure 2. Weld cross section without (solid line) and with (dashed line) magnetic field applied perpendicular to the welding direction (out of plane). Velocity components tangential to the magnetic field do not have an influence on the developing electric currents.

As the resulting Lorentz force decelerates the original melt flow it is expected that the weld pool shape changes from wineglass shape without applied magnetic field to a more uniform V shape of the weld, see Figure 2.

A measure for the extent of flow control by the applied magnetic field is the ratio between the magnetic induced and the viscous drag.

This ratio is called Hartmann number and is defined as follows

$$\mathrm{Ha}^2 = \frac{\sigma(\boldsymbol{B}L)^2}{\eta}.$$

L and η are the weld pool half width and the dynamic viscosity, respectively. Hartmann numbers much larger than unity express that dissipation of kinetic energy in the molten pool occurs mainly by electromagnetic forces instead of viscous damping.

3. Numerical Modeling

The welding velocity was set to 0.5 m/min. The penetration of the laser into the material was around 20 mm. The applied permanent magnets have a quadratic cross section with 50 mm side length. They were mounted on the upper part of the work-piece with around 20 mm overlap and shifted 10 mm to the rear of the weld pool in the simulations.

Three-dimensional heat transfer, fluid dynamics including phase transition and electromagnetic field equations were solved with COMSOL Multiphysics *Non-Isothermal Flow Interface* (NITF) as well as the *Magnetic and Electric Fields Interface* (MEF).

The work-piece is 115 mm in welding direction, 40 mm in lateral direction and 50 mm in vertical direction. Half-symmetry of the weld specimen and surrounding air was exploited resulting in around 900,000 finite elements and around 8.1 millions degrees of freedom by using a segregated solver dividing the variables into the temperature, velocity and pressure variables, turbulence variables, and electrodynamic variables. The convergence criterion was set to 10^{-3}.

Linear finite elements were used for the heat and the Navier-Stokes equations. The Maxwell equations were solved using quadratic elements.

Table 1. Material properties taken from [13, 14]

Material property	Symbol	Value	Unit
Melting temperature	T_{melt}	933	K
Evaporation temperature	T_{evap}	2700	K
Mass density	ρ	2380	kg m^{-3}
Heat capacity	C_p^{eff}	1180	J kg^{-1} K^{-1}
Latent heat of fusion	H_f	$3.97 \cdot 10^5$	J kg^{-1}
Thermal conductivity	λ	91	W m^{-1} K^{-1}
Dynamic viscosity	η	$1.1 \cdot 10^{-3}$	Pa s
Surface tension	γ	0.871	N m^{-1}
Marangoni coefficient	$\partial\gamma/\partial T$	$-1.55 \cdot 10^{-4}$	N m^{-1} K^{-1}
Electrical resistivity	$\rho_{el} = \sigma^{-1}$	$24.77 \cdot 10^{-8}$	Ω m

Main assumptions for the calculations were as follows:

- Steady state simulation.
- Fixed keyhole and weld pool surfaces. The keyhole size was fitted to experimental observations of the weld pool size.
- Turbulent flow pattern.
- The solid-liquid phase change behaviour is modelled by additional source terms in the solid phase.
- Buoyant forces are accounted for using Boussinesq approximation.
- Thermo-physical properties for the numerical simulations were taken from [13, 14]. They are summarized in Figure 3 and Table 1.

The governing equations for the calculations of mass conservation, momentum and energy transport as well as electrodynamic quantities are summarized in the following.

- Mass Conservation

$$\nabla \cdot (\rho \boldsymbol{u}) = 0$$

with mass density ρ.

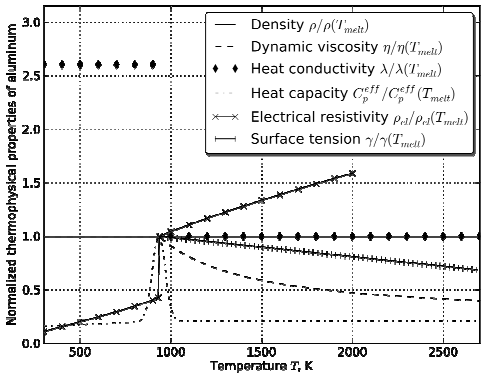


Figure 3. Thermo-physical properties for pure aluminum taken from [13, 14].

- Momentum equation

$$\rho(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \nabla \cdot \left(\eta\left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T\right) - \frac{2}{3}\eta(\nabla \cdot \boldsymbol{u})\boldsymbol{I} \right) + \boldsymbol{F}$$

with pressure p. The source term F holds the buoyant forces, the model for the solidification as well as the electromagnetic force contribution:

$$\boldsymbol{F} = \rho \boldsymbol{g} \beta (T - T_{melt}) - c_1 \frac{(1 - f_L)^2}{f_L^{\,3} + \varepsilon}(\boldsymbol{u} - \boldsymbol{u}_{weld}) + \boldsymbol{j} \times \boldsymbol{B}$$

Here, c_1 and ε are computational constants and f_L denotes the liquid fraction which varies from 0 to 1 in the solidification interval of half width 50 K around the melting temperature.

$$f_L = \begin{cases} 0 & T < T_{sol} \\ \dfrac{T - T_{sol}}{T_{liq} - T_{sol}} & T_{sol} \le T \le T_{liq} \\ 1 & T > T_{liq} \end{cases}.$$

- Energy conservation

$$\rho C_p^{eff} \boldsymbol{u} \cdot \nabla T = \nabla \cdot (\lambda \nabla T)$$

C_p^{eff} denotes an effective heat capacity formulation. T and λ are temperature and heat conductivity. The latent heat amount H_f is normalized around the melting temperature within a temperature range δT of 50 K

$$C_p^{eff} = C_p^0 + \frac{\exp\left[\left(\dfrac{T - T_{melt}}{\delta T}\right)^2\right]}{\sqrt{\pi}\delta T} H_f.$$

The Maxwell equations are valid in stationary form for the magnetic field $\boldsymbol{B}$ and the electric field $\boldsymbol{E}$ as follows:

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{j},$$
$$\nabla \times \boldsymbol{E} = 0.$$

μ_0 is the magnetic permeability of vacuum.

The boundary conditions for the heat and flow equations are summarized in Figure 4. The keyhole domain was cut from the work-piece to allow for liquid metal flow around the keyhole.

At the outer air boundaries, the normal magnetic field component and the normal component of the electric current density vanish. The symmetry plane of the calculation domain was a perfect magnetic conductor with only normal components of the magnetic field. The distance between the permanent magnets was 40 mm. Numerically, the magnetic flux density was applied by a constant boundary condition value at the upper side of the magnets (black surface in Figure 5).

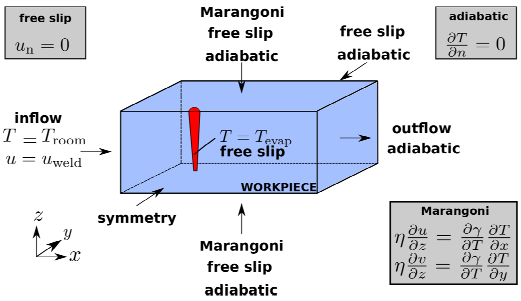


Figure 4. CFD boundary conditions.

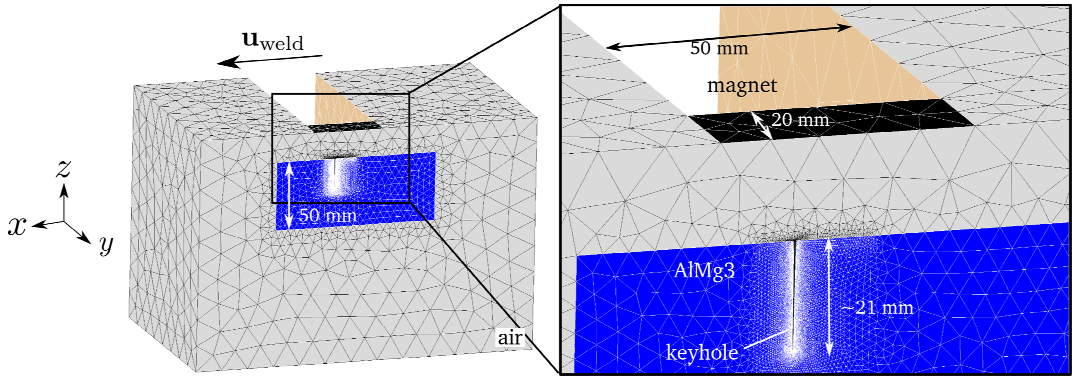


Figure 5. Calculation domain with very fine mesh density around the keyhole due to high thermal gradients.

4. Results

4.1 Reference case without applied magnetic field

Figures 6a and 7a show the symmetry plane view of a characteristic laser beam weld pool for a welding velocity of 0.5 m/min without magnetic fields applied. The colors denote the temperature values and the arrows the velocity field. The solid line represents the melting temperature and the dotted lines temperatures of 50 K below and above melting temperature, respectively, corresponding to the interval where the flow is numerically damped to the processing speed in the solid phase.

A strong vortex at the upper surface is caused by thermo-capillary flow leading to an elongated weld bead near the surface. In the middle and lower part of the weld pool, the flow is mainly dominated by the flow around the keyhole in horizontal direction and additionally by natural convection. The latter causes a convective flow of the melt behind the keyhole.

4.2 Control case with applied magnetic field

The temperature as well as the velocity distributions with different magnetic flux densities applied perpendicular to the welding direction are shown in Figures 6b - 6d and 7b – 7d. The magnetic flux distribution for the case of B=0.5 T is shown in Figure 8.

It yields, that the application of the Lorentz force has a strong influence on both temperature and velocity values. The dominating convexity of the solidification line at the weld pool rear due to the thermo-capillary convection is clearly attenuated and the velocities in the weld pool are much lower compared to the reference case

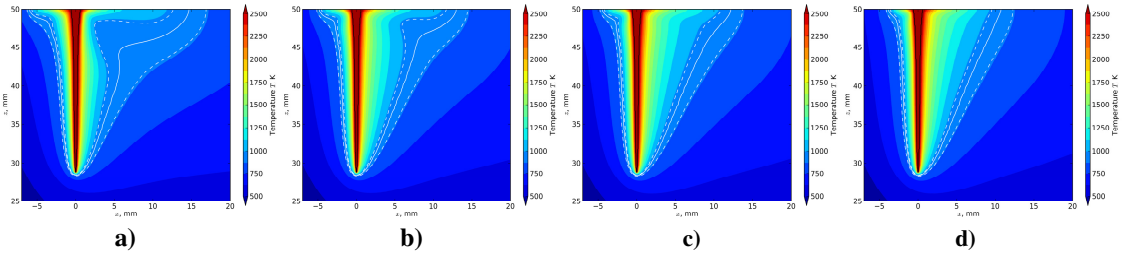


Figure 6. Temperature fields in the symmetry plane for a) B=0, b) B=0.5 T, c) B=1 T, d) B=2 T.

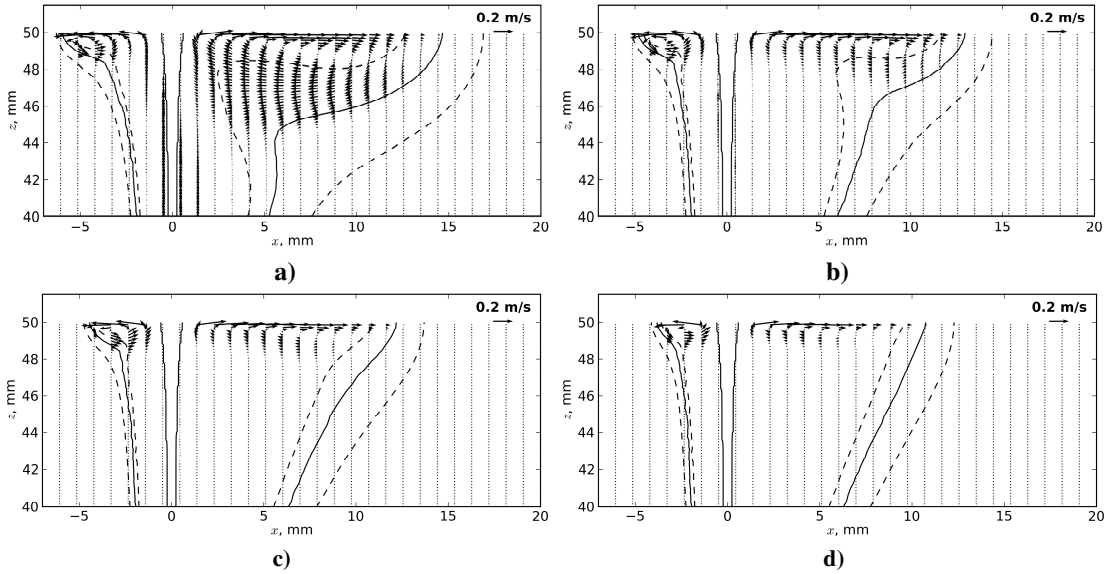


Figure 7. Velocity fields in the symmetry plane for a) B=0, b) B=0.5 T, c) B=1 T, d) B=2 T in the upper region of the weld pool.

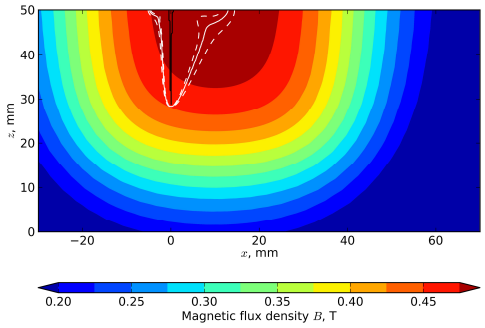


Figure 8. Magnetic flux density distribution in the weld symmetry plane for the case B=0.5 T.

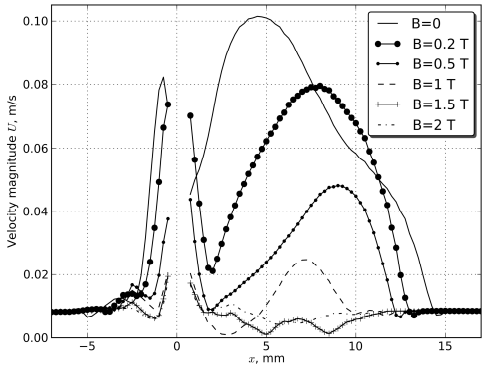


Figure 9. Velocity distributions 2 mm below the upper surface along the welding direction. Position 0 in x-direction corresponds to the laser spot position.

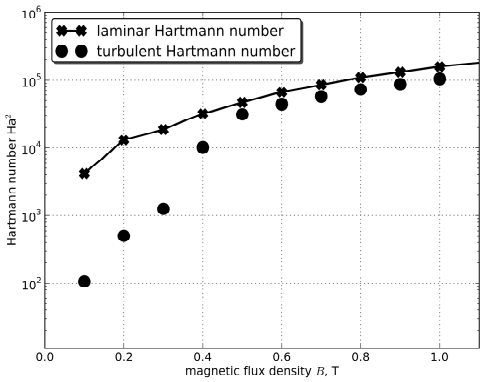


Figure 10. Hartmann number calculated based on the half weld pool width for increasing magnetic flux densities with and without considering the turbulent viscosity contribution.

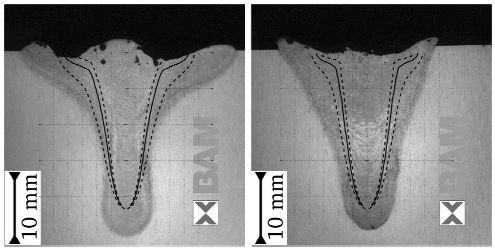


Figure 11. Comparison between weld macro sections perpendicular to the welding direction from simulation and experiment for the reference case (left) without applied Lorentz forces and the case B=0.5 T (right).

without magnetic fields applied. Thus the potential for severe spattering and melt ejections is heavily degraded.

It becomes also clear, that the influence of the thermo-capillary vortex at the upper surface is the more limited to a thin surface layer the higher the applied magnetic flux density is.

The velocity distribution for increasing magnetic field at 2 mm below the upper surface is shown in Figure 9. The velocities slow down the larger the applied magnetic field and its subsequent Lorentz forces are. Velocities below the welding speed indicate a flow reversal from recirculation mode (Figure 2a) to a flow parallel aligned to the welding direction (Figure 2d).

The process of making the flow laminar can be seen from the Hartmann number calculated at a point 5 mm below the surface and 5 mm behind the keyhole shown in Figure 10, with and without consideration of the turbulent viscosity being added to the dynamic viscosity material property value. It yields, that beginning from a point where the magnetic field is sufficient to eliminate turbulence for the most part (B=0.5 T), both curves are similar meaning that turbulence has a vanishing influence on the flow behaviour.

Figure 11 shows the experimental comparison to welding experiments with a fibre laser with 16 kW output power. The laser focus was 4 mm below the surface and the welding velocity was 0.5 m/min. The experiments show the same tendency as the simulations predict, namely a narrowing of the weld cross section and a lesser convexity of the solidification line as well as a smoother surface topology due to electromagnetic damping. Differences between simulation and experiment can be justified by the fact that the welding was done with AlMg3 alloy whereas the calculation was accomplished with pure aluminium due to a lack of availability of

high-temperature material data in the literature. The material properties of both pure aluminium and AlMg3 are slightly different.

5. Conclusions

The influence of a steady state magnetic field perpendicular to the welding direction was investigated numerically and experimentally.

The induced magnetic drag component clearly lowers the influence of the Marangoni stresses at the upper surface which is due to the temperature-dependent surface tension.

It is shown, that the application of a magnetic flux density of 0.5 T with a corresponding Hartmann number around 10^4 leads to the local development of a laminar flow regime due to the magnetic induced drag component that is directed against the original melt velocity thus reducing turbulence.

Globally, the higher the magnetic field applied, the more the convexity of the melting isotherm due to thermo-capillary flow is diminished in both the longitudinal as well as the transversal cross sections which is advantageous to avoid excessive mechanical stresses in the welds after cooling-down. Lowering the melt velocities is also favorable to prevent strong spattering and melt ejections driven by the dynamics in the weld pool.

6. References

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